CPU

File: cpu.kicad_sch

Radio

File: radio.kicad_sch

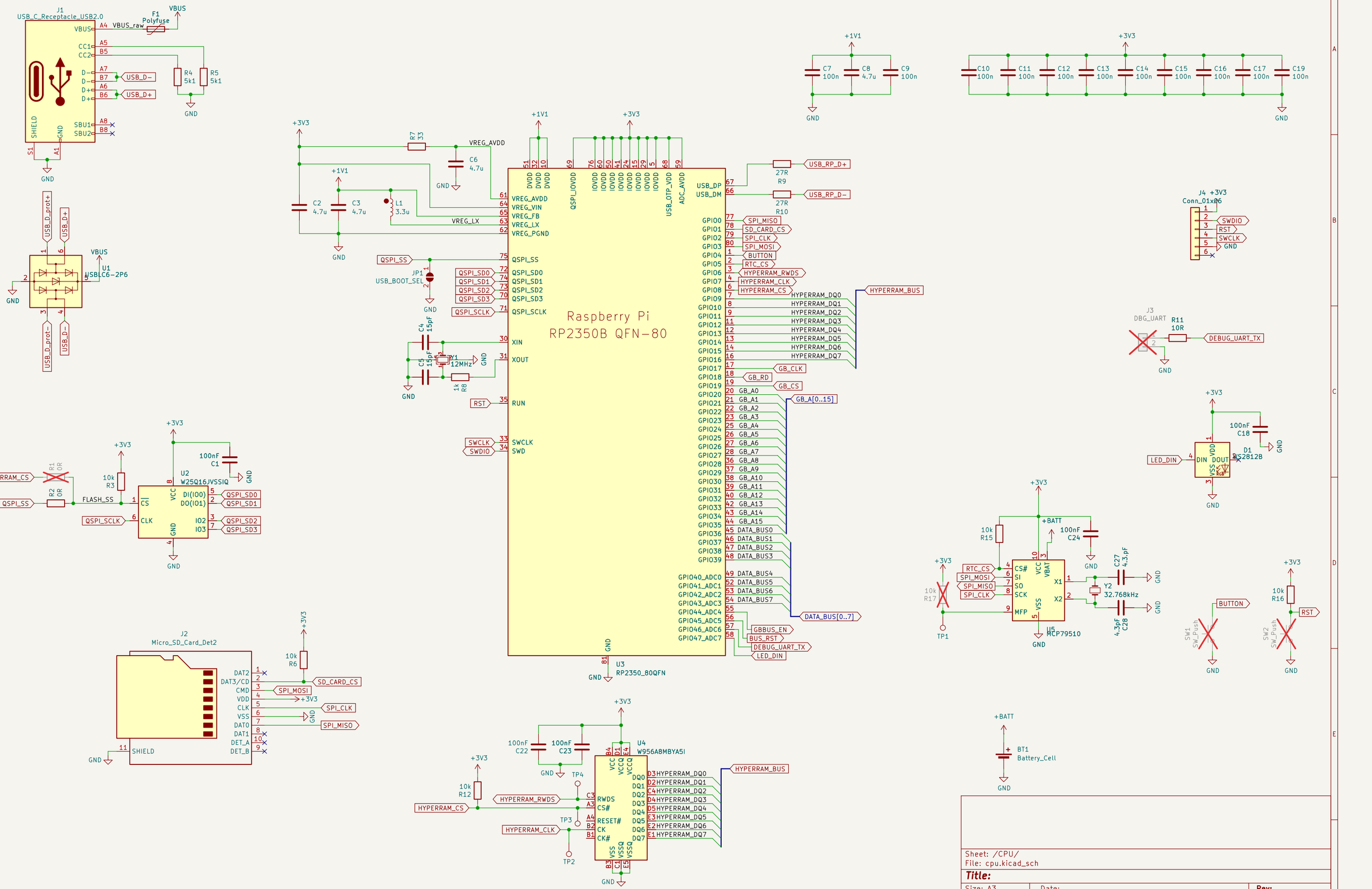
Power_GBBus

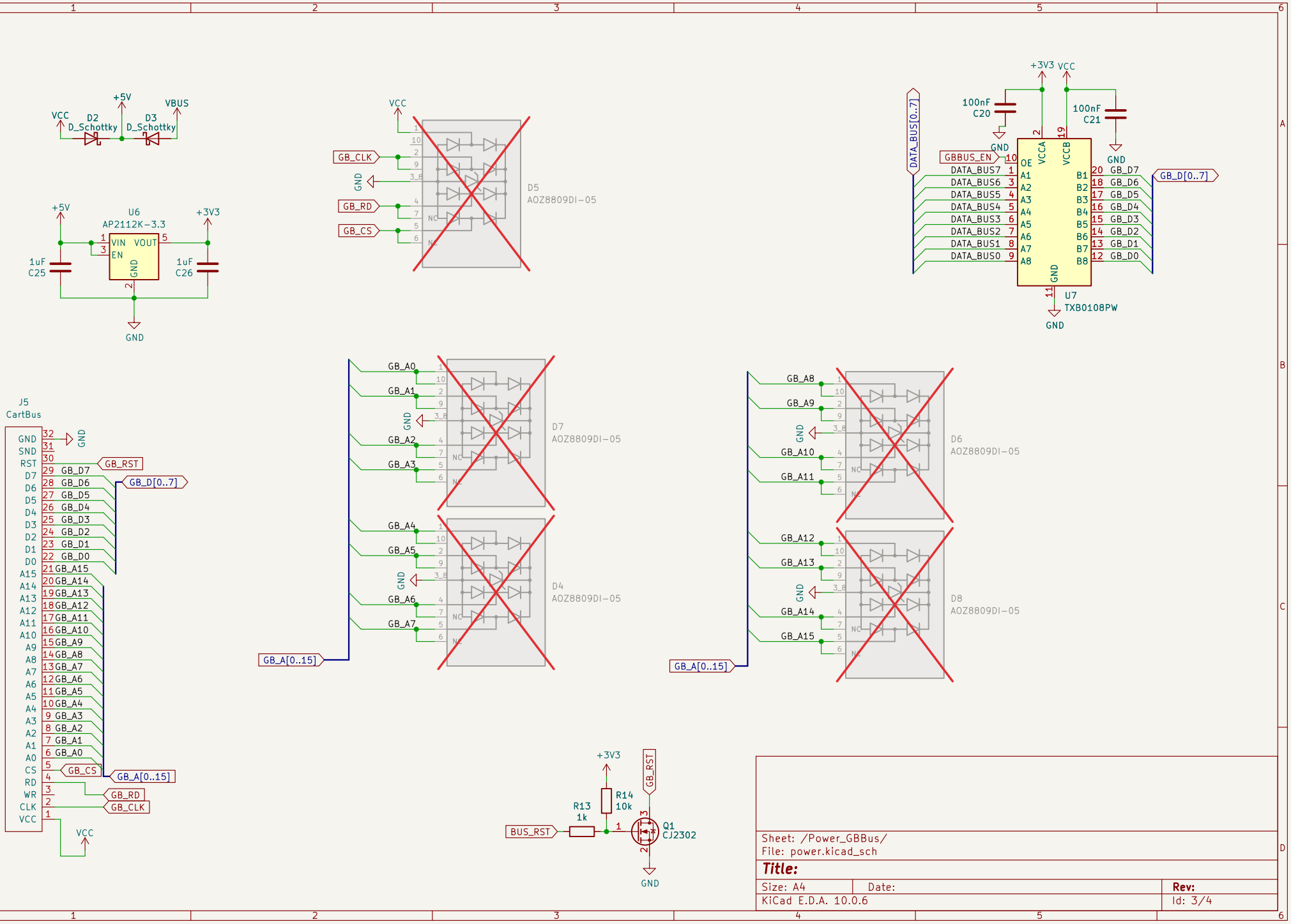
File: power.kicad_sch

Sheet: /
File: GameboyCartridgeV2.1.kicad_sch

Title:

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J1
USB_C_Receptacle_USB2.0
F1
Polyfuse
VBUS
VBUS_raw
SHIELD
GND
CC1
CC2
D−
D+
SBU1
SBU2
S1
A1
A4
A5
B5
A7
B7
A6
B6
A8
B8
USB_D−
USB_D+
R4
5k1
R5
5k1
USB_D_prot+
USB_D_prot−
U1
USBLC6−2P6
+1V1
+3V3
C7
100n
C8
4.7u
C9
100n
C10
100n
C11
100n
C12
100n
C13
100n
C14
100n
C15
100n
C16
100n
C17
100n
C19
100n
R7
33
VREG_AVDD
C6
4.7u
C2
4.7u
C3
4.7u
L1
3.3u
VREG_LX
VREG_VIN
VREG_FB
VREG_PGND
QSPI_SS
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SCLK
JP1
USB_BOOT_SEL
C4
15pF
C5
15pF
Y1
12MHz
1k
R8
XIN
XOUT
RUN
RST
SWCLK
SWD
SWDIO
DVDD
IOVDD
QSPI_IOVDD
USB_OTP_VDD
ADC_AVDD
USB_DP
USB_DM
27R
R9
27R
R10
USB_RP_D+
USB_RP_D−
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
GPIO24
GPIO25
GPIO26
GPIO27
GPIO28
GPIO29
GPIO30
GPIO31
GPIO32
GPIO33
GPIO34
GPIO35
GPIO36
GPIO37
GPIO38
GPIO39
GPIO40_ADC0
GPIO41_ADC1
GPIO42_ADC2
GPIO43_ADC3
GPIO44_ADC4
GPIO45_ADC5
GPIO46_ADC6
GPIO47_ADC7
SPI_MISO
SD_CARD_CS
SPI_CLK
SPI_MOSI
BUTTON
RTC_CS
HYPERRAM_RWDS
HYPERRAM_CLK
HYPERRAM_CS
HYPERRAM_DQ0
HYPERRAM_DQ1
HYPERRAM_DQ2
HYPERRAM_DQ3
HYPERRAM_DQ4
HYPERRAM_DQ5
HYPERRAM_DQ6
HYPERRAM_DQ7
HYPERRAM_BUS
GB_CLK
GB_RD
GB_CS
GB_A0
GB_A1
GB_A2
GB_A3
GB_A4
GB_A5
GB_A6
GB_A7
GB_A8
GB_A9
GB_A10
GB_A11
GB_A12
GB_A13
GB_A14
GB_A15
GB_A[0..15]
DATA_BUS0
DATA_BUS1
DATA_BUS2
DATA_BUS3
DATA_BUS4
DATA_BUS5
DATA_BUS6
DATA_BUS7
DATA_BUS[0..7]
GBBUS_EN
BUS_RST
DEBUG_UART_TX
LED_DIN
U3
RP2350_80QFN
Raspberry Pi
RP2350B QFN−80
J4 +3V3
Conn_01x06
SWDIO
RST
SWCLK
J3
DBG_UART
R11
10R
DEBUG_UART_TX
100nF
C18
D1
WS2812B
LED_DIN
J2
Micro_SD_Card_Det2
10k
R6
SD_CARD_CS
SPI_MOSI
SPI_CLK
SPI_MISO
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DET_A
DET_B
SHIELD
U2
W25Q16JVSSIQ
HYPERRAM_CS
QSPI_SS
FLASH_SS
QSPI_SCLK
R1
0R
R2
0R
10k
R3
100nF
C1
U4
W956A8MBYA5I
HYPERRAM_RWDS
HYPERRAM_CLK
TP2
TP3
TP4
10k
R12
100nF
C22
100nF
C23
+BATT
BT1
Battery_Cell
10k
R15
100nF
C24
C27
4.3pF
Y2
32.768kHz
C28
4.3pF
U5
MCP79510
TP1
10k
R17
BUTTON
SW1
SW_Push
10k
R16
RST
SW2
SW_Push
Sheet: /CPU/
File: cpu.kicad_sch
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VCC
D2
D_Schottky
+5V
D3
D_Schottky
VBUS
+5V
U6
AP2112K-3.3
+3V3
VIN
VOUT
EN
GND
1uF
C25
1uF
C26
GND
VCC
GB_CLK
GND
GB_RD
GB_CS
D5
AOZ8809DI-05
+3V3
VCC
100nF
C20
100nF
C21
GND
DATA_BUS[0..7]
GBBUS_EN
DATA_BUS7
DATA_BUS6
DATA_BUS5
DATA_BUS4
DATA_BUS3
DATA_BUS2
DATA_BUS1
DATA_BUS0
OE
VCCA
VCCB
GND
GB_D7
GB_D6
GB_D5
GB_D4
GB_D3
GB_D2
GB_D1
GB_D0
GB_D[0..7]
U7
TXB0108PW
GND
J5
CartBus
GND
SND
RST
D7
D6
D5
D4
D3
D2
D1
D0
A15
A14
A13
A12
A11
A10
A9
A8
A7
A6
A5
A4
A3
A2
A1
A0
CS
RD
WR
CLK
VCC
GB_RST
GB_D[0..7]
GB_CS
GB_A[0..15]
GB_RD
GB_CLK
VCC
GB_A0
GB_A1
GB_A2
GB_A3
GB_A4
GB_A5
GB_A6
GB_A7
GND
D7
AOZ8809DI-05
D4
AOZ8809DI-05
GB_A[0..15]
GB_A8
GB_A9
GB_A10
GB_A11
GB_A12
GB_A13
GB_A14
GB_A15
GND
D6
AOZ8809DI-05
D8
AOZ8809DI-05
GB_A[0..15]
+3V3
R14
10k
R13
1k
GB_RST
BUS_RST
Q1
CJ2302
GND
Sheet: /Power_GBBus/
File: power.kicad_sch
Title:
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Date:
Rev:
KiCad E.D.A. 10.0.6
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U101
TPS22810RLR
R101
5.1k 1%
R102
2.0k 1%
C101
1uF 25V X5R
C102
1uF 25V X5R
C103
1uF 25V X5R
R115
100R 1%
U102
TPS2553DBVR
R103
180k 1%
R104
100k 1%
C104
1uF 25V X5R
C105
330uF 6.3V 25mOhm 6TPE330MAP
C106
330uF 6.3V 25mOhm 6TPE330MAP
C107
330uF 6.3V 25mOhm 6TPE330MAP
C108
10uF 10V X5R
C109
10uF 10V X5R
U108
TLV3012BIDBVR
R123
1.5M 1%
R124
1.0M 1%
R125
1.2M 1%
C121
10nF 50V X7R
C122
100nF 16V X7R
U103
TPS63802DLAR
L101
0.47uH 5.5A 26mOhm DFE201612E-R47M
R109
511k 1%
R110
91k 1%
C110
22uF 6.3V X5R
R108
10k 1%
R111
10k 1%
C113
100nF 16V X7R
U104
ESP32-S3-MINI-1U
C111
10uF 10V X5R
C112
100nF 16V X7R
R112
10k 1%
R113
10k 1%
R114
10k 1%
R116
680k 1%
C114
2.2nF 50V X7R
R117
200k 1%
R118
680k 1%
R119
200k 1%
C115
2.2nF 50V X7R
U105
SN74LVC1G125DBVR
C116
100nF 16V X7R
U106
SN74LVC3G34DCUR
U107
CH334P
R122
33R 1%
Y101
12MHz 10pF 10ppm
C118
1uF 25V X5R
C119
1uF 25V X5R
C120
100nF 16V X7R
R120
22R 1%
R121
22R 1%
U109
TLV431BIDBZR
Q101
SIA975DJ-T1-GE3
R126
27.0k 0.1%
R127
10.0k 0.1%
R128
680 1%
R129
680 1%
C123
470pF 50V C0G
R130
330k 1%
D101
1N4148WS
D102
1N4148WS
D103
1N4148WS
C124
100nF 16V X7R
R131
470k 1%
D104
1N4148WS
Q102
2N7002
R132
1k 1%
PWR_FLAG

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Sheet: /Radio/
File: radio.kicad_sch

Title: Radio + console-safe power stage

Size: A1	Date:	Rev: B0
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